Programming Assignment #3

Decorate 3D models with lighting effects and textures

Due:

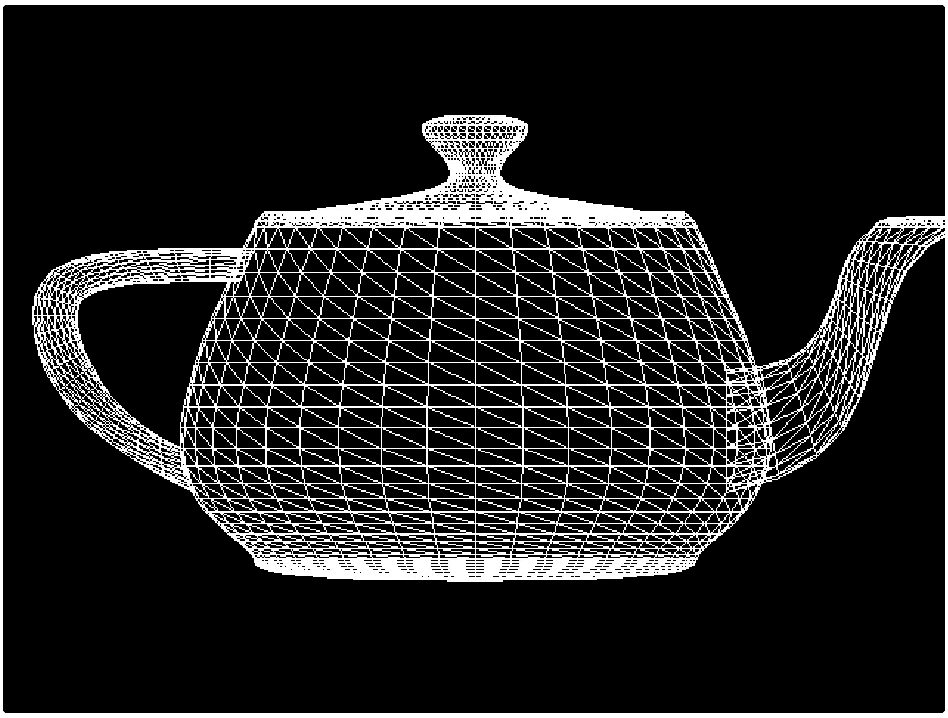
Please check on ETL

Submission:

Please zip all files (program + supplemental material) then upload it on the ETL

Specification:

- Basics
  - **[1pt]** Load and draw a provided rock model (".obj" format) with **wireframe** rendering
    - File: https://imolab-my.sharepoint.com/:u:/g/personal/jungdam_imo_snu_ac_kr/EV_HKY2YKPdPqW85WPFiffwB8ohHqlfHRx9T6dnpiBGFfA?e=DUEV15
    - Obj description:
      - https://www.youtube.com/watch?v=iClme2zsg3I
      - cs.cmu.edu/~mbz/personal/graphics/obj.html
    - Wireframe
      - Wireframe rendering means that vertices and edges are drawn only with constant color (e.g., black or blue)
      - Example



- Local Illumination Models
    - **[3pt]** Step1
        - Render the rock w/o textures
            - Implement Phong / Gouraud illumination model using vertex and pixel shaders
            - Recommended parameters:
                - K_a = (0.1, 0.1, 0.1)
                - K_d = (0.5, 0.5, 0.5)
                - K_s = (0.8, 0.8, 0.8)
                - n = 4 or 8
            - Compare Phong/Gouraud shading (triangle interpolation)
    - **[2pt]** Step2
        - Render the rock w/ textures
            - Use the provided textures to decorate your model
                - K_a = Free_rock_Mixed_AO.jpg x "Free_rock_Base_Color.jpg"
                    - Load both files and multiply them to get the final K_a
                - K_d = Free_rock_Base_Color.jpg
                - K_s = Free_rock_Specular.jpg
                - n ← Free_rock_Roughness.jpg
                    - The value should be inverted because 'n' represents 'shineness' in Phong illumination model. Given a roughness value x, its shineness n can be obtained by n = 1 / (a x + b), where a,b are tunable parameters.
            - Compare your results with the ground-truth file below (due to light setup your results might not be perfectly the same but try to make it as similar as possible)
                - Free_rock.blend
                - To see the result, you need to install Blender(https://www.blender.org/download/)

- **[2pt]** Step3
  - Implement normal mapping using the file below (use Tangent-space normal method)
    - Free_rock_Normal_OpenGL.jpg
- Artistic score
  - Create your custom scene which are aesthetically pleasing or technically illustrative
    - Use any **FREE** models in (https://www.turbosquid.com/)
    - Multiple objects can be used in one scene
    - Your own rendering alg. implemented above should be used to render the scene
    - Submit 3 best screenshots
    - DO NOT add animations on your scene. The scene should be